

Long-range order and spontaneous symmetry breaking in quantum spin systems and hard-core bosons on a lattice

part 3 LRO and SSB in the ground states of quantum spin systems with discrete symmetry

***Advanced Topics in
Statistical Physics
by Hal Tasaki***

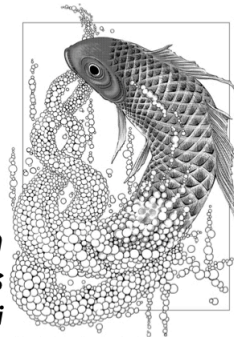


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§ Ising model under a transverse magnetic field

a simple theoretical example of quantum many-body system

Hamiltonian

$$\hat{H}_L = - \sum_{\{u,v\} \in \mathcal{B}_L} \hat{S}_u^{(z)} \hat{S}_v^{(z)} - \lambda \sum_{u \in \Lambda_L} \hat{S}_u^{(x)} \quad (1) \quad \lambda \geq 0 \text{ a fixed parameter}$$

$\hat{S}_L^{(x)} = \sum_{u \in \Lambda_L} \hat{S}_u^{(x)}$

We focus on the ground state(s) of $\hat{H}_L$ for large L in any $d=1,2,\dots$

Symmetry

($\hat{T}_w$: translation $\hat{T}_w^\dagger \hat{H}_L \hat{T}_w = \hat{H}_L$ for any w (2))

$$\hat{U} = \hat{U}_L^{(x,\pi)} \quad (3) \quad \hat{U}^\dagger \hat{H}_L \hat{U} = \hat{H}_L \quad (4)$$

$$\hat{U}^2 = \hat{I} \quad (5) \quad \mathbb{Z}_2 \text{ symmetry}$$

order operator

$$\hat{O}_L^{(z)} = \sum_u \hat{S}_u^{(z)} \quad (6) \quad \hat{U}^\dagger \hat{O}_L^{(z)} \hat{U} = -\hat{O}_L^{(z)} \quad (7)$$

▶ trivial case with $\lambda = 0$

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easy because $[\hat{H}_L, \hat{O}_L^{(2)}] = 0$ (1)

doubly degenerate exact ground states

$$|\Phi_L^\uparrow\rangle = \bigotimes_{u \in \Lambda_L} |\uparrow\rangle_u \quad (2)$$

$$|\Phi_L^\downarrow\rangle = \bigotimes_{u \in \Lambda_L} |\downarrow\rangle_u \quad (3)$$

$$\hat{H}_L |\Phi_L^\uparrow\rangle = E_L^{\text{GS}} |\Phi_L^\uparrow\rangle \quad (4)$$

$$\hat{H}_L |\Phi_L^\downarrow\rangle = E_L^{\text{GS}} |\Phi_L^\downarrow\rangle \quad (5)$$

$$\text{with } E_L^{\text{GS}} = -\frac{1}{4} |\mathcal{B}_L| \quad (6)$$

each of the ground state breaks the $\mathbb{Z}_2$ symmetry explicitly

$$\hat{O}_L^{(2)} |\Phi_L^\uparrow\rangle = \frac{L^d}{2} |\Phi_L^\uparrow\rangle \quad (7)$$

$$\hat{O}_L^{(2)} |\Phi_L^\downarrow\rangle = -\frac{L^d}{2} |\Phi_L^\downarrow\rangle \quad (8)$$

exact ground states $|\Phi_L^\uparrow\rangle, |\Phi_L^\downarrow\rangle$ are physical ground states
with LRO and symmetry breaking

► case with $\lambda > 0$

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↪ nontrivial and interesting because $[\hat{H}_L, \hat{O}_L^{(2)}] \neq 0$ (1)

theorem for any L , the ground state of $\hat{H}_L$ is unique and written as
 $|\Phi_L^{GS}\rangle = \sum_{\sigma \in \mathcal{S}} \alpha_\sigma |\sigma\rangle$ (2) with $\alpha_\sigma > 0$ for any $\sigma \in \mathcal{S}$

(proof: • $\langle \sigma | \hat{H}_L | \tau \rangle \in \mathbb{R}$, $\langle \sigma | \hat{H}_L | \tau \rangle \leq 0$ if $\sigma \neq \tau$
• any $\sigma, \tau \in \mathcal{S}$ are "connected" via nonzero $\langle \sigma' | \hat{H}_L | \sigma'' \rangle$)

Perron-Frobenius theorem (for real symmetric matrices) implies the theorem

uniqueness + symmetry of $\hat{H}_L \rightarrow \hat{U} |\Phi_L^{GS}\rangle = C_L |\Phi_L^{GS}\rangle$ with $|C_L| = 1$ (3)

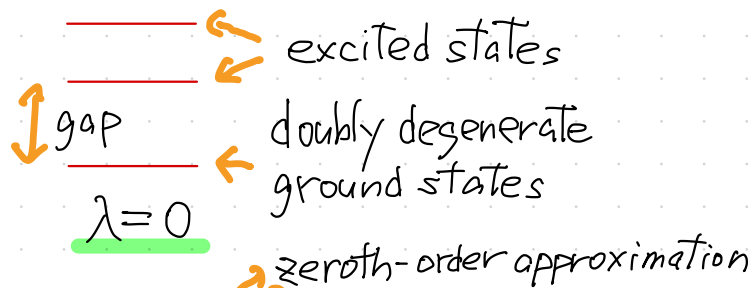
$$\langle \Phi_L^{GS} | \hat{O}_L^{(2)} | \Phi_L^{GS} \rangle = \langle \Phi_L^{GS} | \hat{U}^\dagger \hat{O}_L^{(2)} \hat{U} | \Phi_L^{GS} \rangle = - \langle \Phi_L^{GS} | \hat{O}_L^{(2)} | \Phi_L^{GS} \rangle \quad (4)$$

$$\langle \Phi_L^{GS} | \hat{O}_L^{(2)} | \Phi_L^{GS} \rangle = 0 \quad (4)$$

NO symmetry breaking

case with $0 < \lambda \ll 1$ perturbation of the trivial model at $\lambda=0$

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$$E_L^{1st} - E_L^{gs} \sim e^{-cL^d}$$

exponentially small splitting

$$|\Phi_L^{gs}\rangle \simeq \frac{1}{\sqrt{2}} \{ |\Phi_L^\uparrow\rangle + |\Phi_L^\downarrow\rangle \}$$

(1) unique exact ground state

$$|\Phi_L^{1st}\rangle \simeq \frac{1}{\sqrt{2}} \{ |\Phi_L^\uparrow\rangle - |\Phi_L^\downarrow\rangle \}$$

(2) low-lying excited state

apparently Schrödinger's cats **unphysical**



they must collapse into $|\Phi_L^\uparrow\rangle$ or $|\Phi_L^\downarrow\rangle$

since $(\hat{O}_L^{(z)})^2 |\Phi_L^\uparrow\rangle = \left(\frac{L^d}{2}\right)^2 |\Phi_L^\uparrow\rangle$ (3) $(\hat{O}_L^{(z)})^2 |\Phi_L^\downarrow\rangle = \left(\frac{L^d}{2}\right)^2 |\Phi_L^\downarrow\rangle$ (4)

$$\langle \Phi_L^{gs} | \left(\frac{\hat{O}_L^{(z)}}{L^d} \right)^2 | \Phi_L^{gs} \rangle \simeq \frac{1}{2} \left\{ \langle \Phi_L^\uparrow | \left(\frac{\hat{O}_L^{(z)}}{L^d} \right)^2 | \Phi_L^\uparrow \rangle + \langle \Phi_L^\downarrow | \left(\frac{\hat{O}_L^{(z)}}{L^d} \right)^2 | \Phi_L^\downarrow \rangle \right\} = \frac{1}{4}$$

long-range order!

(5)

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recalling $\langle \Phi_L^{GS} | \hat{O}_L^{(2)} | \Phi_L^{GS} \rangle = 0$ (1) (no symmetry breaking)

we see

$$\sqrt{\langle \Phi_L^{GS} | \left(\frac{\hat{O}_L^{(2)}}{L^d} \right)^2 | \Phi_L^{GS} \rangle - \left(\langle \Phi_L^{GS} | \frac{\hat{O}_L^{(2)}}{L^d} | \Phi_L^{GS} \rangle \right)^2} \simeq \frac{1}{2} \quad (2)$$

fluctuation of $\hat{O}_L^{(2)} / L^d$ in $|\Phi_L^{GS}\rangle$

The density of a macroscopic quantity has nonzero fluctuation

the exact ground state $|\Phi_L^{GS}\rangle$ with LRO but without SSB is unphysical as a state of a macroscopic system

what are physical ground states?

two ground states because $\mathbb{Z}_2$ symmetry is broken

$|\Phi_L^\uparrow\rangle$ and $|\Phi_L^\downarrow\rangle$ are macroscopic ground states

with both LRO and SSB (zeroth-order approximation for $0 < \lambda \ll 1$)

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▣ macroscopic ground state general definition (only in this mini-course)

$\hat{H}_L$ translation-invariant Hamiltonian on Λ_L
with short-range interactions

ground state energy density $\epsilon^{\text{GS}} = \lim_{L \uparrow \infty} \frac{E_L^{\text{GS}}}{L^d} \quad (1)$

$\hat{T}_w |\Psi_L\rangle = \text{const.} |\Psi_L\rangle \quad (2)$

if $|\Psi_L\rangle$ is translation invariant and satisfies

$$\lim_{L \uparrow \infty} \frac{1}{L^d} \langle \Psi_L | \hat{H}_L | \Psi_L \rangle = \epsilon^{\text{GS}} \quad (3)$$

we say $|\Psi_L\rangle$ (for large L) is a macroscopic ground state

infinite volume state $\omega(\cdot)$ defined by $\omega(\hat{A}) = \lim_{L \uparrow \infty} \langle \Psi_L | \hat{A} | \Psi_L \rangle$
is a ground state in the sense of the operator algebraic formulation
of quantum spin systems. $\rightarrow$ Appendix A of Koma-Tasaki 1994

back to the model with $0 < \lambda \ll 1$

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$|\Phi_L^{GS}\rangle, |\Phi_L^{1st}\rangle$ macroscopic ground states with LRO but without SSB **unphysical**

$|\Phi_L^\uparrow\rangle, |\Phi_L^\downarrow\rangle$ macroscopic ground states with LRO and SSB **physical**
(zeroth-order approximation for $0 < \lambda \ll 1$)

hints for general theories

$$\begin{aligned}\hat{O}_L^{(2)} |\Phi_L^{GS}\rangle &\simeq \hat{O}_L^{(2)} \frac{1}{\sqrt{2}} \{ |\Phi_L^\uparrow\rangle + |\Phi_L^\downarrow\rangle \} \\ &\simeq \frac{L^d}{2} \frac{1}{\sqrt{2}} \{ |\Phi_L^\uparrow\rangle - |\Phi_L^\downarrow\rangle \} \simeq \frac{L^d}{2} |\Phi_L^{1st}\rangle \quad (1)\end{aligned}$$

this suggests

$$|\Phi_L^\uparrow\rangle \simeq \frac{1}{\sqrt{2}} \{ |\Phi_L^{GS}\rangle + |\Phi_L^{1st}\rangle \} \simeq \frac{1}{\sqrt{2}} \left\{ |\Phi_L^{GS}\rangle + \frac{\hat{O}_L^{(2)} |\Phi_L^{GS}\rangle}{\|\hat{O}_L^{(2)} |\Phi_L^{GS}\rangle\|} \right\} \quad (2)$$

$$|\Phi_L^\downarrow\rangle \simeq \frac{1}{\sqrt{2}} \{ |\Phi_L^{GS}\rangle - |\Phi_L^{1st}\rangle \} \simeq \frac{1}{\sqrt{2}} \left\{ |\Phi_L^{GS}\rangle - \frac{\hat{O}_L^{(2)} |\Phi_L^{GS}\rangle}{\|\hat{O}_L^{(2)} |\Phi_L^{GS}\rangle\|} \right\} \quad (3)$$

§ general theories for low-lying states and SSB

Horsch, von der Linden 1988, Kaplan, Horsch, von der Linden 1989

setting

$\hat{H}_L$ translation-invariant Hamiltonian on Λ_L with short-range interactions

$\hat{O}_L = \sum_{u \in \Lambda_L} \hat{O}_u$ (1) order operator ($\hat{O}_u = \hat{O}_u^\dagger$ acts only on u , $\hat{O}_u = \hat{T}_u \hat{O}_0 \hat{T}_u^\dagger$)

$|\Phi_L^{GS}\rangle$ exact translation-invariant ground state of $\hat{H}_L$

E_L^{GS} ground state energy and its density $\epsilon^{GS} = \lim_{L \uparrow \infty} \frac{E_L^{GS}}{L^d}$ (2)

assumptions

$$\langle \Phi_L^{GS} | \left(\frac{\hat{O}_L}{L^d} \right)^2 | \Phi_L^{GS} \rangle \geq q_0 > 0 \quad (3) \quad \langle \Phi_L^{GS} | \hat{O}_u \hat{O}_v | \Phi_L^{GS} \rangle \gtrsim q_0 \text{ for large } |u-v|$$

there is LRO

$$\langle \Phi_L^{GS} | \left(\frac{\hat{O}_L}{L^d} \right)^m | \Phi_L^{GS} \rangle = 0 \text{ for } m=1,3 \quad (4) \quad \text{but no SSB}$$

▷ low-lying states

define $|\Gamma_L\rangle = \frac{\hat{O}_L |\Phi_L^{GS}\rangle}{\|\hat{O}_L |\Phi_L^{GS}\rangle\|}$ (1)

see p.7-(1)

Theorem (Horsch, von der Linden 1988)

$|\Gamma_L\rangle$ is orthogonal to $|\Phi_L^{GS}\rangle$ there is a constant $C > 0$ s.t.

$$0 \leq \langle \Gamma_L | \hat{H}_L | \Gamma_L \rangle - E_L^{GS} \leq \frac{C}{L^d} \quad (2)$$

Corollary there is an energy eigenstate $|\Psi_L\rangle$ s.t. $\langle \Phi_L^{GS} | \Psi_L \rangle = 0$

with eigenvalue E_L^1 satisfying $0 \leq E_L^1 - E_L^{GS} \leq \frac{C}{L^d}$ (3)

→ (proof: easy)

then $\lim_{L \rightarrow \infty} \frac{1}{L^d} \langle \Gamma_L | \hat{H}_L | \Gamma_L \rangle = \lim_{L \rightarrow \infty} \frac{1}{L^d} \langle \Psi_L | \hat{H} | \Psi_L \rangle = E^{GS}$ (4)

$|\Gamma_L\rangle$ and $|\Psi_L\rangle$ are macroscopic ground states

choose $|\Psi_L\rangle$ to be translation invariant

proof of theorem

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$$|\Gamma_L\rangle = \frac{\hat{O}_L |\Phi_L^{GS}\rangle}{\|\hat{O}_L |\Phi_L^{GS}\rangle\|} \quad (1)$$

$$\langle \Phi_L^{GS} | \Gamma_L \rangle = \frac{\langle \Phi_L^{GS} | \hat{O}_L | \Phi_L^{GS} \rangle}{\|\hat{O}_L |\Phi_L^{GS}\rangle\|} = 0 \quad (2)$$

p8-(4)

we write $\langle \dots \rangle = \langle \Phi_L^{GS} | \dots | \Phi_L^{GS} \rangle$ (3)

$$\langle \Gamma_L | \hat{H}_L | \Gamma_L \rangle = \frac{\langle \Phi_L^{GS} | \hat{O}_L \hat{H}_L \hat{O}_L | \Phi_L^{GS} \rangle}{\langle \Phi_L^{GS} | \hat{O}_L \hat{O}_L | \Phi_L^{GS} \rangle} = \frac{\langle OHO \rangle}{\langle O^2 \rangle} \quad (4)$$

$\hat{O}_L^\dagger = \hat{O}_L$

$$\langle \Gamma_L | \hat{H}_L | \Gamma_L \rangle - E_L^{GS} = \frac{\langle OHO \rangle - E_L^{GS} \langle O^2 \rangle}{\langle O^2 \rangle}$$

we shall evaluate
this double commutator!

$$= \frac{\langle OHO \rangle - \frac{1}{2} \langle O^2 H \rangle - \frac{1}{2} \langle HO^2 \rangle}{\langle O^2 \rangle} = \frac{\langle [O, [H, O]] \rangle}{2 \langle O^2 \rangle} \quad (5)$$

$$\hat{H}_L = \sum_{u \in \Lambda_L} \hat{h}_u \quad (1) \quad \hat{h}_u \text{ acts only on sites } v \text{ s.t. } |u-v| \leq r_0$$

$$\hat{O}_L = \sum_{u \in \Lambda_L} \hat{O}_u \quad (2) \quad \hat{O}_u \text{ acts only on site } u$$

$$[\hat{O}_L, [\hat{H}_L, \hat{O}_L]] = \sum_{u, v, w \in \Lambda_L} [\hat{O}_u, [\hat{h}_v, \hat{O}_w]] = \sum_{v \in \Lambda_L} \sum_{\substack{u, w \\ (|u-v| \leq r_0) \\ (|w-v| \leq r_0)}} [\hat{O}_u, [\hat{h}_v, \hat{O}_w]] \quad (3)$$

$$\|[\hat{O}_L, [\hat{H}_L, \hat{O}_L]]\| \leq \sum_{v \in \Lambda_L} \sum_{\substack{u, w \\ (|u-v| \leq r_0) \\ (|w-v| \leq r_0)}} \|[\hat{O}_u, [\hat{h}_v, \hat{O}_w]]\| \leq A L^d \quad (4)$$

operator norm (see next page) $\rightarrow \leq 4 \|\hat{h}_0\| \|\hat{O}_0\|^2$

$$\langle [\hat{O}, [\hat{H}, \hat{O}]] \rangle \leq A L^d \quad (5)$$

$$p8-(3) \rightarrow \langle \hat{O}^2 \rangle \geq q_0 L^{2d} \quad (6) \leftarrow \text{LRO!!}$$

$$\langle \Gamma | \hat{H}_L | \Gamma \rangle - E_L^{\text{GS}} = \frac{\langle [[\hat{O}, [\hat{H}, \hat{O}]]] \rangle}{2 \langle \hat{O}^2 \rangle} \leq \frac{A L^d}{2 q_0 L^{2d}} = \frac{C}{L^d} \quad (7) //$$

operator norm

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$\mathcal{H}$ finite dimensional Hilbert space

$\hat{A}$ operator on $\mathcal{H}$

$$\|\hat{A}\| = \max_{|\Phi\rangle \in \mathcal{H}} \|\hat{A}|\Phi\rangle\| \quad (1)$$

$(\|\Phi\|=1)$

$$\|\hat{A} + \hat{B}\| \leq \|\hat{A}\| + \|\hat{B}\| \quad (2)$$

$$\|\hat{A}\hat{B}\| \leq \|\hat{A}\| \|\hat{B}\| \quad (3)$$

$$|\langle \Phi | \hat{A} | \Phi \rangle| \leq \|\hat{A}\| \langle \Phi | \Phi \rangle \quad (4)$$

if $\hat{A} = \hat{A}^\dagger$ with eigenvalues $a_1, \dots, a_D$

$$\|\hat{A}\| = \max_j |a_j| \quad (5)$$

macroscopic ground states with LRO and SSB

define $|\square_L^\pm\rangle = \frac{1}{\sqrt{2}} \{ |\Phi_L^{GS}\rangle \pm |\Gamma_L\rangle \}$ → p7-(2),(3) (1)

clearly $\|\square_L^\pm\| = 1$ (2) $\langle \square_L^+ | \square_L^- \rangle = 0$ (3)

note also $\langle \square_L^\pm | \hat{H}_L | \square_L^\pm \rangle = \frac{1}{2} \{ \langle \Phi_L^{GS} | \hat{H}_L | \Phi_L^{GS} \rangle + \langle \Gamma_L | \hat{H}_L | \Gamma_L \rangle \}$
 $\leq \frac{1}{2} \{ E_L^{GS} + E_L^{GS} + C/L^d \} = E_L^{GS} + C/(2L^d)$

$|\square_L^\pm\rangle$ are macroscopic ground states

$$\langle \square_L^+ | \hat{O}_L | \square_L^+ \rangle = \frac{1}{2} \left(\langle \Phi_L^{GS} | + \frac{\langle \Phi_L^{GS} | \hat{O}_L}{\|\hat{O}_L | \Phi_L^{GS} \rangle} \right) \hat{O}_L \left(|\Phi_L^{GS}\rangle + \frac{\hat{O}_L | \Phi_L^{GS} \rangle}{\|\hat{O}_L | \Phi_L^{GS} \rangle} \right)$$

p8-(4) →

$$= \frac{\langle \Phi_L^{GS} | (\hat{O}_L)^2 | \Phi_L^{GS} \rangle}{\|\hat{O}_L | \Phi_L^{GS} \rangle} = \sqrt{\langle \Phi_L^{GS} | (\hat{O}_L)^2 | \Phi_L^{GS} \rangle} \geq \sqrt{q_0} L^d$$
← p8-(3)

$$\pm \langle \square_L^\pm | \frac{\hat{O}_L}{L^d} | \square_L^\pm \rangle \geq \sqrt{q_0}$$

$|\square_L^\pm\rangle$ exhibits (spontaneous) symmetry breaking!

$|\square_L^\pm\rangle$ may or may not be a physical state with small fluctuation
 See Appendix B, Komatsu & Tasaki (1994)

▶ Griffiths-type theorem

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$$\hat{H}_L^h = \hat{H}_L - h \hat{O}_L \quad (1) \quad |\Phi_L^{GS,h}\rangle \text{ an exact ground state of } \hat{H}_L^h$$

↘ symmetry breaking field

Theorem (Kaplan, Horsch, von der Linden 1989)

$$\lim_{h \downarrow 0} \lim_{L \uparrow \infty} \langle \Phi_L^{GS,h} | \frac{\hat{O}_L}{L^d} | \Phi_L^{GS,h} \rangle \geq \sqrt{c_0} > 0 \quad (2)$$

part 2 - p.4

infinitesimal symmetry breaking field leads to a ground state with SSB

proof $\langle \mathbb{E}_L^+ | \hat{H}_L^h | \mathbb{E}_L^+ \rangle \geq \langle \Phi_L^{GS,h} | \hat{H}_L^h | \Phi_L^{GS,h} \rangle \quad (3)$

divide by $h L^d$

↗ definition of a ground state!

$$\begin{aligned} \langle \Phi_L^{GS,h} | \frac{\hat{O}_L}{L^d} | \Phi_L^{GS,h} \rangle &\geq \langle \mathbb{E}_L^+ | \frac{\hat{O}_L}{L^d} | \mathbb{E}_L^+ \rangle + \frac{1}{h L^d} \{ \langle \Phi_L^{GS,h} | \hat{H}_L | \Phi_L^{GS,h} \rangle - \langle \mathbb{E}_L^+ | \hat{H}_L | \mathbb{E}_L^+ \rangle \} \\ &\geq \sqrt{c_0} + \frac{1}{h L^d} \{ E_L^{GS} - (E_L^{GS} + \frac{c}{2 L^d}) \} \quad (4) \end{aligned}$$

↘ $\rightarrow 0 \text{ as } L \rightarrow \infty$